

Progress Report — Week 8

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Report return date: 26 June 2026

Reporting period

19 June 2026 – 25 June 2026

Hours worked

Date	Day	Hours	Activity summary
19 Jun 2026	Fri	8	UML class and component diagrams for testbed architecture
22 Jun 2026	Mon	8	CI performance causal diagram (07_detailed_uml_ci_performance.mmd)
23 Jun 2026	Tue	8	CONCEPTUAL_MODEL.md — theory chain and variable glossary
24 Jun 2026	Wed	8	ER-style and sprint-loop sequence diagrams
25 Jun 2026	Thu	8	One-page presentation diagram for thesis/demo use
Total		40	

Work completed

- Created UML and conceptual diagrams in docs/diagrams/: architecture, sprint loop, CI subconstruct map, effectiveness model.
 - Wrote docs/CONCEPTUAL_MODEL.md explaining theory chain from inputs → CI → decision quality → sprint outcomes → effectiveness.
 - Produced detailed CI performance diagram showing links between agile variables and CI subconstructs.
 - Ensured diagrams match current code structure after recent model refinements.
 - Prepared visual materials suitable for thesis presentation and conference demo.
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Connection to learning objectives

- Applied modelling and technical communication skills to make complex systems understandable.
 - Worked independently on documentation while ensuring alignment with implemented code.
 - Supported future academic dissemination through clear visual artifacts.
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Challenges and how they were handled

Diagrams became outdated quickly as the model evolved. I established a rule to update diagrams in the same work session as related code changes.

Plan for next week (26 Jun – 2 Jul 2026)

- Write `UML_CI_PERFORMANCE_ANALYSIS.md` with paper citations and weight rationale.
 - Add agile variable columns and explain simulation assumptions vs empirical coefficients.
 - Export analysis document to HTML and PDF.
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Student signature: Arslan Sultan — Date: 26 June 2026

Supervisor acknowledgment: Juan Garbajosa